

- 2 Species diversity is often used as a simple way to compare the biodiversity of different regions of the world.

A student read that species diversity decreases with increasing distance from the equator. The latitude at the equator is 0° . Latitudes increase north and south of the equator.

The student selected data from a database which holds records of the annual bird count in North America.

This annual count is carried out each year during one day in the winter. Volunteers count all the species of birds that they observe in each location. The area of each location is the same size.

The student found data from 17 sample locations within the Delmarva Peninsula on the east coast of the USA for 2005.

Fig. 2.1 shows the 17 locations within the Delmarva Peninsula.

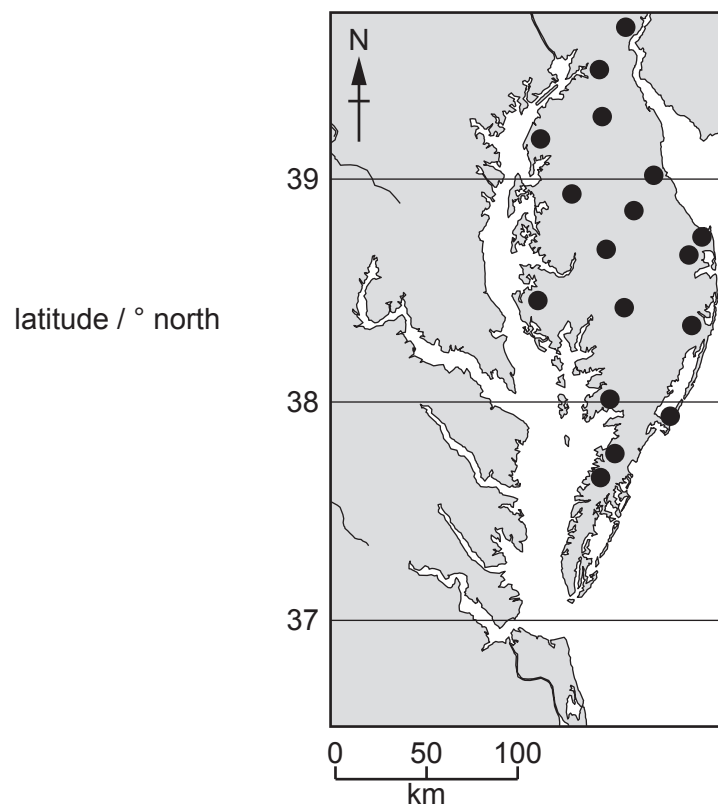


Fig. 2.1

The student used the data to see if there was any relationship between the latitudes of the sample locations and the number of species recorded at each location.

The data are shown in Fig. 2.2.

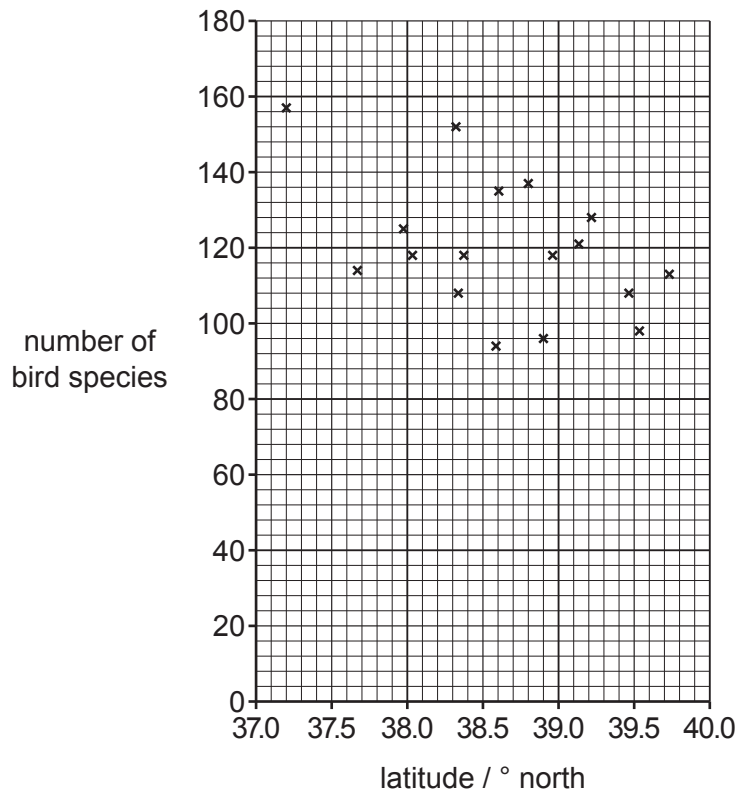


Fig. 2.2

(a) Describe the trend shown by the scatter graph in Fig. 2.2.

.....

.....

..... [1]

(b) The student analysed the data using Spearman's rank correlation test.

(i) State **three** reasons why this statistical test is a suitable way to analyse the data about species diversity in Fig. 2.2.

.....

.....

.....

.....

.....

..... [3]

(ii) State the null hypothesis for this investigation.

.....

..... [1]

- (c) The student was investigating the statement 'species diversity decreases with increasing distance from the equator'.

Discuss the limitations of the sampling method and of the data collected in supporting this statement.

.....

.....

.....

.....

.....

.....

..... [3]

Turn over for question 2(d).

Small birds such as the ruby-crowned kinglet, *Regulus calendula*, are found throughout North America.

Fig. 2.3 shows a ruby-crowned kinglet caught in a mist net. Mist nets are used in surveys of bird populations.



Fig. 2.3

- (d) Describe how the mark-release-recapture method can be used to estimate the population size of small birds, such as the ruby-crowned kinglet.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 15]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.